Computational Lithography Solutions for Non-Immersion ArF (193nm) Excimer Laser Deep Ultraviolet Machines for Chemically Amplified Resists with Bottom Anti-Reflective Coating

Executive Program in Semiconductor Manufacturing - IIT Delhi

Tejas Beedkar
Sushma Shetty
Nitesh Mahawar
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Index

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Chapter 3 - ArF Laser Cabinet

3.1 ArF Excimer Laser Cabinet

An Argon Fluoride (ArF) excimer laser is a specific type of deep ultraviolet (DUV) gas-discharge laser. It is the primary light source utilized in advanced immersion lithography systems to print nanoscale semiconductor circuits.

The term "excimer" is short for *excited dimer*. Under a high-voltage electrical discharge, noble gas atoms (Argon) and reactive halogen atoms (Fluorine) are forced together to form a temporary, unstable molecular bound state (ArF^*). When this short-lived molecule collapses back into its ground state, it dissociates and releases a highly energetic photon in the deep ultraviolet spectrum.

This laser cannot be used for Lithography without sigificnt conditioning. The before and after parameters/characteristics are listed below

3.1.1 — Beam Conditioning Chain (Raw Excimer Output → Illuminator)

These are all the stages that the Laser cabinet will implement.

Stage / Subsystem	Component	Function
Line Narrowing Module (LNM)	Beam Expander (prism pair or grating)	Expands the beam spatially before the dispersive element to increase angular resolution.
Line Narrowing Module (LNM)	Echelle / Littrow Diffraction Grating	Primary dispersive element. High-order diffraction selects a narrow spectral slice.

Stage / Subsystem	Component	Function
Line Narrowing Module (LNM)	Etalon (Fabry–Pérot, sometimes two in series)	Reflective interference cavity stacked with the grating for a combined bandwidth of ~0.1–0.3 pm. The key spectral line-narrowing element.
Line Narrowing Module (LNM)	Tuning Actuators (PZT or stepper motors)	Adjust grating angle and etalon gap to center the output wavelength precisely (e.g., 193.3683 nm for ArF).
Metrology / Wavelength Module	Wavelength & Bandwidth Monitor (etalon + grating spectrometer + photodetector)	Measures λ and bandwidth shot-by-shot; provides feedback to the tuning actuators and dose control.
Pulse Conditioning	Pulse Stretcher (optical delay / OPuS-type beam-splitter network)	Splits and recombines the pulse over delay paths to lengthen TIS, lowering peak power and reducing optics damage/compaction.
Energy Control	Variable Attenuator (rotating plates / pulsed energy control module)	Trims pulse energy to the dose target without changing laser drive; first stage of dose regulation.
Beam Delivery	Beam Delivery Unit / Beam Expander	Relays the beam from laser to scanner over distance; expander shapes size and divergence for the illuminator input.
Beam Delivery	Coherence Scrambler / Spatial Coherence Reducer	Reduces spatial coherence to suppress speckle and improve illumination uniformity downstream.
Spectral Cleanup (KrF systems)	DUV / Spectral Purity Filter	Blocks unwanted out-of-band emission (e.g., red ASE in KrF) so only the working wavelength reaches the illuminator. Not applicable to 193nm ArF lasers

3.1.2 Key Beam Characteristics: Source vs. Illuminator-Stage Delivery with Component → Parameter Mapping

This table gives a comparision of the laser parametres from production and as it is fed to the illuminator. The laser cabinet conditions the laser to achive the required characteristics.

The state that achieves each parameter is also mentioned.

Characteristic	Raw Source Value	Illuminator-Stage Value	Conditioning Component Responsible
Central Wavelength (λ)	193.36 nm (free-running)	193.368 nm, stabilized	Grating + etalon (selection); tuning actuators (centering)

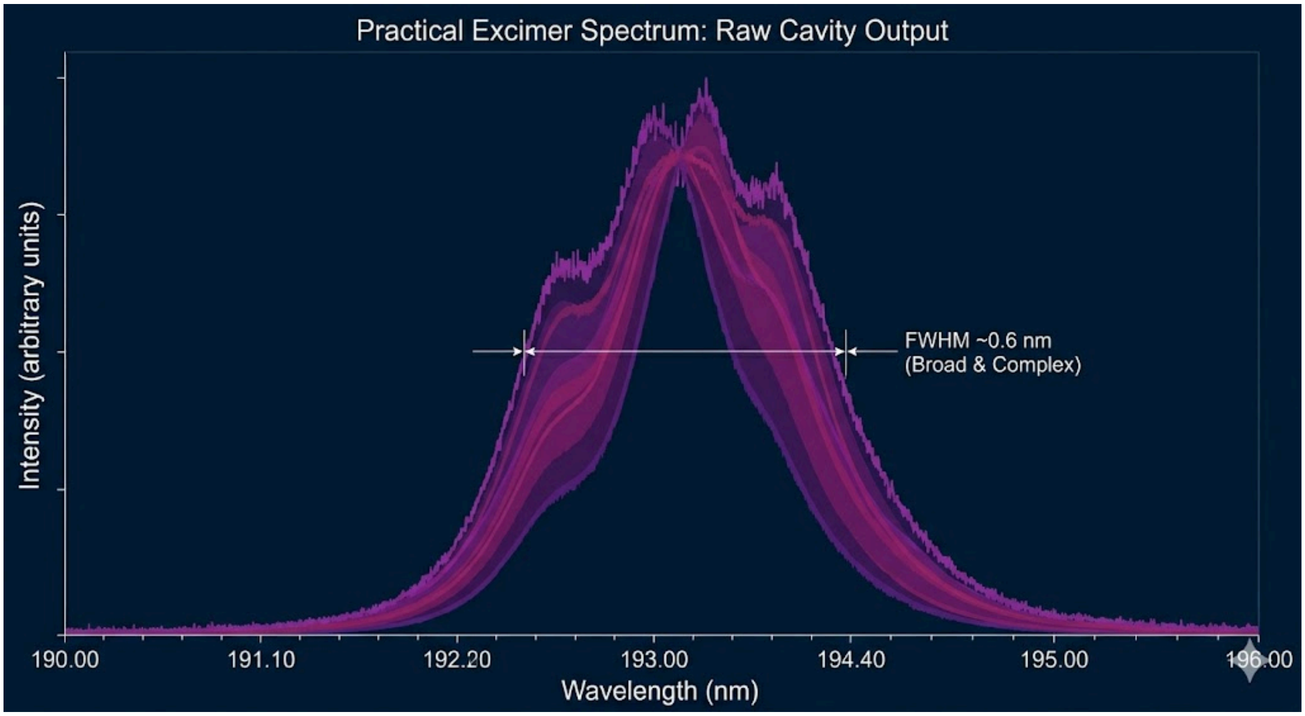
Characteristic	Raw Source Value	Illuminator-Stage Value	Conditioning Component Responsible
Wavelength Stability	Uncontrolled / drifts	< 0.02–0.05 pm (±)	Wavelength monitor + tuning actuators (closed-loop feedback)
Bandwidth (FWHM)	~300–500 pm (raw)	< 0.15–0.3 pm (narrowed)	Beam expander + grating + etalon (the LNM as a whole)
Spectral Purity (E95)	Not specified / very broad	< 0.3–0.5 pm	LNM (narrowing) + DUV/spectral-purity filter (out-of-band rejection)
Pulse Duration	~20–60 ns	~50–120 ns (stretched, TIS)	Pulse stretcher
Pulse Repetition Rate	4–6 kHz	4–6 kHz (unchanged)	None — set by laser discharge electronics
Average Output Power	40–90 W	Attenuated to dose target	Variable attenuator / energy control module
Dose Stability (per field)	Pulse-to-pulse varies (~few %)	< 0.3–0.5% (3σ)	Energy monitor + attenuator (closed-loop dose control)
Illumination Uniformity	Non-uniform raw profile	< 0.5–1.0% across slit	Coherence scrambler (speckle reduction) + downstream homogenizer
Field Shape	Raw beam (~few mm × mm)	Rectangular slit (~26 mm × 5–8 mm)	Beam delivery / beam expander (size & divergence shaping)
Beam Divergence / Pointing	Larger divergence, raw jitter	Stabilized, low jitter	Beam delivery unit (relay optics + pointing stabilization)

3.2 Excimer Spectrum (Raw Cavity Output and Ideal)

3.2.1 Practical Excimer Spectrum (Raw Cavity Output)

This diagram visualizes the actual, raw light generated inside the laser cavity before any narrowing. This is the output that is described in the table as "spectrally broad" and unusable.

While still centered around 193 nm, this peak is a complex "mountain range" rather than a sharp spike. It is visibly much fatter. It is composed of multiple overlapping bands of light (purple/magenta hues). The Full Width at Half Maximum (FWHM) is shown conceptually as ~0.6 nm, which is hundreds of times wider than the ideal spectrum. This broad width is what would cause catastrophic blurring in the projection lens.



The laser operating point is the specific combination of electrical and gas conditions at which the excimer laser produces stable, repeatable, on-specification output. It is not a single number — it is a set of interdependent parameters that must all be held simultaneously:

Parameter	What it controls	Consequence if moved off-point

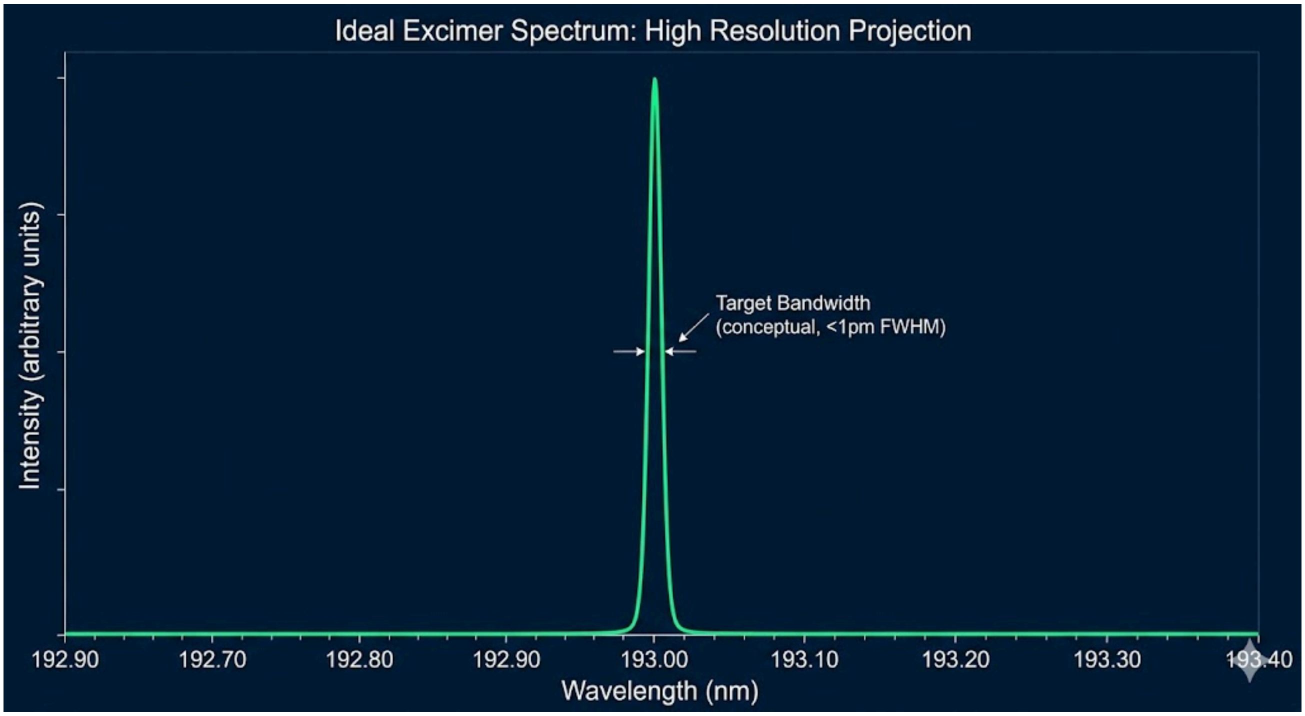
|Discharge voltage |Energy deposited into the plasma per pulse |Too low → weak pulse, unstable discharge · Too high → arcing, gas contamination, electrode erosion |Gas pressure |Density of the Kr+F_2 or Ar+F_2 mixture |Changes gain bandwidth, center wavelength, and pulse shape| Gas mixture ratio |Ratio of halogen (F_2) to noble gas (Kr or Ar) to buffer (Ne or He) |F_2 depletion over time shifts gain spectrum and reduces output energy| |Pulse repetition rate |How often the discharge fires (1–4 kHz) |Too fast → gas does not recover between pulses → bandwidth widens | |Discharge timing / pulse width |Duration of the electrical discharge |Affects plasma temperature and thus spectral line shape |Gas temperature |Via heat exchangers on the chamber |Thermal expansion changes gas density → shifts operating pressure|

The laser is always operated at the its peak performance operating point. All laser conditioning is done is the downstream stages of the DUV piepline.

3.2.2 Ideal Excimer Spectrum for DUV Photolithography

This diagram visualizes the requirement after all spectral treatment—this is what the lithography system needs to avoid chromatic aberration.

The peak is centered precisely at the target wavelength (e.g., 193.0 nm for ArF), and the crucial feature is its absolute purity: the bandwidth is incredibly narrow, conceptualized here as being in the picometer range (e.g., <1 pm FWHM). It is a single, sharp, intense vertical line (a near-delta function) on a dark background.

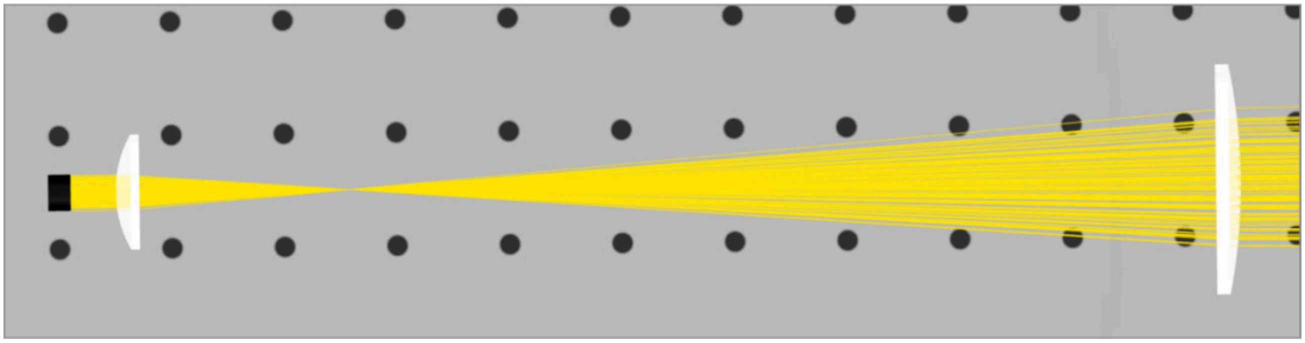


3.3 Line Narrowing Module (LNM) / Line Narrowing Package (LNP)

3.3.1 Beam Expander (Prism Pair)

The Explanation: A beam expander does exactly what its name suggests: it takes the narrow, raw laser beam coming out of the excimer cavity and stretches it out spatially, usually in one dimension (making it wider horizontally). This is typically done using a series of specialized prisms.

Why it's needed: The beam expander does not filter colors itself. Instead, it prepares the beam for the diffraction grating. A grating's ability to separate colors (its "angular resolution") depends entirely on how many microscopic grooves the laser light hits. By expanding the beam to be much wider, the light covers significantly more grooves on the grating, massively increasing the precision of the spectral filtering that follows.



```
===============================================================================
                        STAGE 1: BEFORE BEAM EXPANDER
===============================================================================

  [ SPATIAL FOOTPRINT ]                    [ SPECTRAL COMPONENT ]
   (Narrow Slit Shape)                       (Broad, Raw Spectrum)

       Width: ~3-5 mm                                Intensity
          |<->|                                          |         /\
       __ +---+                                          |        /  \
        ^ |///|                                          |       /    \
 Height | |///|                                          |    /\/      \/\
 ~15 mm | |///|                                          |   /            \
        v |///|                                          |  /              \
       -- +---+                                          | /                \
                                                         +---------------------- Wavelen
                                                              <-- ~0.6 nm -->
                                                                  (FWHM)
===============================================================================
```

```
===============================================================================
                         STAGE 1: AFTER BEAM EXPANDER
===============================================================================

  [ SPATIAL FOOTPRINT ]                    [ SPECTRAL COMPONENT ]
  (Expanded for Grating)            (Spectrum remains UNCHANGED here)

                      Width: ~100+ mm                     Intensity
          |<--------------------------------------->|         |         /\
       __ +-----------------------------------------+         |        /  \
        ^ |/////////////////////////////////////////|         |       /    \
 Height | |/////////////////////////////////////////|         |    /\/      \/\
 ~15 mm | |/////////////////////////////////////////|         |   /            \
        v |/////////////////////////////////////////|         |  /              \
       -- +-----------------------------------------+         | /                \
                                                              +-------------------
                                                                  <-- ~0.6 nm -->
                                                                      (FWHM)
===============================================================================
```

Parameter Changed: Spatial Beam Width (Note: The beam expander does not alter the spectrum; it alters the physical geometry of the beam).

- Before (Narrow Spatial Profile): The beam is a concentrated, narrow rectangle of light exiting the raw laser cavity.
- After (Expanded Spatial Profile): The beam is physically stretched horizontally, maintaining its height but becoming much wider, preparing to blanket the entire surface of the diffraction grating.

NOTE- Expanded beam is collimated before being processed in the diffractor.

https://www.omnicalculator.com/physics/laser-beam-expander

https://www.silloptics.de/en/service/sill-technical-guide/laser-optics/beam-expanders

https://www.3doptix.com/use-cases/laser-beam-expanders-i/

https://precisionlaserscanning.com/2016/03/beam-expander-is-critical-for-beam-shaping/

3.3.2 Echelle / Littrow Diffraction Grating

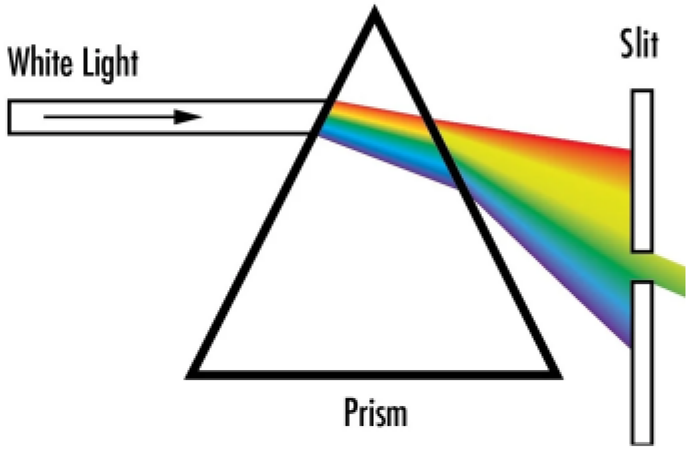
A diffraction grating is an optical component covered in thousands of perfectly parallel, microscopic grooves. Like a prism, it separates incoming light into its component colors (wavelengths), but a grating does this through diffraction and interference rather than refraction, making it far more powerful.

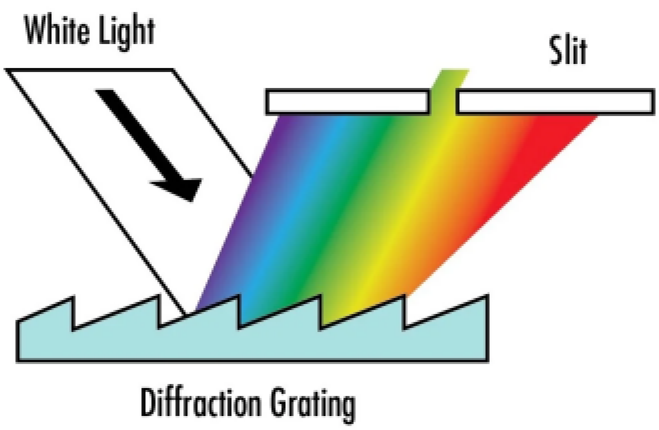
Littrow Configuration: This refers to the specific angle at which the grating is tilted. The beam hits the grating and diffracts backwards along the exact same path it came from.

Echelle: This means the grating is optimized to use high "orders" of diffraction, which spreads the wavelengths out at incredibly extreme angles.

Function: This is the "coarse" filter. It takes the messy raw spectrum and spatially spreads out the different microscopic shades of UV light. By placing an aperture (a tiny slit) in the path of the returning light, only the specific target wavelength (and a small band around it) is allowed to pass back into the laser cavity. The rest of the raw spectrum is physically blocked.

While dispersion prisms separate wavelengths through refraction (top), diffraction gratings instead separate wavelengths through diffraction because of their surface structure (bottom).





Parameter Changed: Spectral Bandwidth (Coarse Narrowing)

- Before (Raw Spectrum): The broad, messy spectrum (~0.6 nm FWHM) full of unwanted satellite wavelengths.
- After (Grating-Narrowed Spectrum): A significantly refined spectrum. The broad "mountain" of light is shaved down to a much narrower spike (e.g., ~10 to 50 picometers FWHM). However, it is still not narrow enough for advanced lithography.

```
================================================================================
                   BEFORE GRATING: RAW EXCIMER SPECTRUM
================================================================================

Intensity
  1.0 |                              /\
      |                             /  \
  0.8 |                            /    \
      |                         /\/      \/\
  0.6 |                        /            \     <-- Wide, messy "mountain"
      |                       /              \
  0.4 |----------------------/----------------\------------------ Half Maximum
      |                     /                  \
  0.2 |                    /                    \
      |        ___________/                      \___________
  0.0 +-------+----------+----------+----------+----------+------- Wavelength
           192.5      192.8       193.0      193.2      193.5      (nm)
                                 (Target)

                            |<-----------------|
                                FWHM ≈ 0.6 nm
                           (Catastrophic for lenses)
================================================================================
```

```
================================================================================
                 AFTER GRATING: COARSELY NARROWED SPECTRUM
================================================================================

Intensity
  1.0 |                               /\
      |                               ||
  0.8 |                               ||
      |                               ||
  0.6 |                               ||           <-- Unwanted light physically
      |                               ||               blocked/diffracted away
  0.4 |-------------------------------||------------------------ Half Maximum
      |                               ||
  0.2 |                               ||
      |                               ||
  0.0 +-------+----------+----------+-||-+----------+----------+------- Wavelength
           192.5      192.8      193.0      193.2      193.5      (nm)
                                (Target)

                                   |<->|
                                FWHM ≈ 30 pm
                                 (0.03 nm)
                         (Much better, but still requires
                          Etalon for final fine-tuning)
================================================================================
```

Why the grating is essential to make the Etalon comb work

The etalon alone would pass multiple comb teeth — every green peak within the gain bandwidth of the excimer (~0.5 nm wide) would transmit. Without the grating pre-selecting a ~1–5 pm window first, the laser would simultaneously lase at many comb teeth. The grating kills all neighbouring teeth by ensuring only one sits within the selected band. The etalon then sharpens that single surviving tooth to sub-pm width.

Ref -

https://www.lighttrans.com/fileadmin/shared/UseCases/Application_UC_Highly%20Efficient%20Polariz

https://www.lighttrans.com/use-cases/application/analysis-and-design-of-highly-efficient-polarization-independent-transmission-gratings.html?utm_source=Newsletter&utm_campaign=NL+KW9.2+%282%29

https://www.lighttrans.com/newsletter-area/grating-systems-in-littrow-configuration.html

https://www.mdpi.com/2076-3417/12/21/11042

https://www.researchgate.net/figure/Possible-Echelle-grating-optical-configurations-Echelle-grating-has-right_fig1_251233810

https://www.rp-photonics.com/diffraction_gratings.html

https://www.sciencedirect.com/topics/physics-and-astronomy/echelle-gratings

https://www.oejournal.org/oee/en/supplement/726db385-9a46-421c-8365-5ce124f3f749

https://www.edmundoptics.com/knowledge-center/application-notes/optics/all-about-diffraction-gratings/

https://www.sciencedirect.com/science/article/abs/pii/S0030399225020663

https://opg.optica.org/viewmedia.cfm?r=1&rwjcode=josa&uri=josa-39-7-522&html=true

3.3.3 Etalon (Fabry–Pérot)

The grating takes the FWHM down from roughly 600 picometers (0.6 nm) to roughly 30 picometers. While this is a massive improvement—stripping away 95% of the unwanted spectral garbage—it is still considered "coarse" narrowing in the context of advanced lithography. The beam must now pass through the Etalon (the fine filter) to crush that 30 pm spike down to the final target of < 0.3 pm

The etalon acts as the ultimate microscopic comb. It takes the output from the grating and uses optical resonance to shave it down to the absolute ideal width required for the projection lenses.

A reflective interference cavity (sometimes two etalons in series) stacked with the grating to achieve a combined bandwidth of ~0.1–0.3 pm. This is the key spectral line-narrowing element.

The Explanation:

The etalon is a pair of parallel, partially-reflective flat surfaces — typically a solid fused-silica or CaF_2 plate with precision-coated faces. It works on the principle of optical resonance - Light enters, bounces multiple times between the two surfaces, and the reflections interfere with each other. Only wavelengths that satisfy the constructive-interference condition survive and transmit through; all others are reflected back.

- Wavelengths that do not perfectly "fit" the gap between the mirrors destructively interfere with each other and cancel out.
- Only the exact wavelengths that perfectly resonate within the gap constructively interfere and are transmitted through the other side.

The Etalon acts like an incredibly precise comb. When stacked with the diffraction grating, the Etalon shaves the final picometers off the spectrum, achieving the ultra-pure <0.3 pm FWHM target.

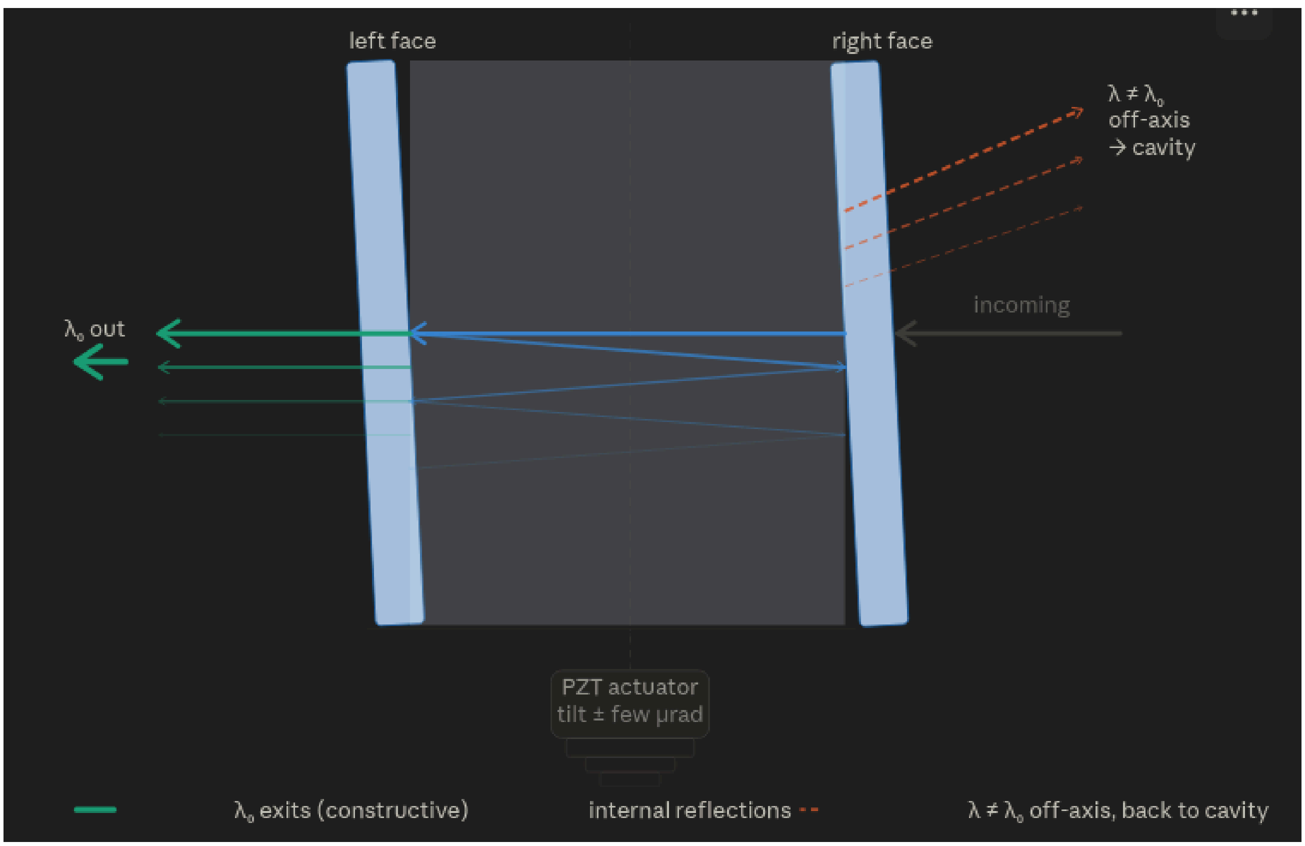
It is purely a filter. The etalon selects which wavelength wins the gain competition; it does not shift, convert, or alter any wavelength. The photons that exit are exactly the same wavelength as they were inside the gain medium. No frequency conversion occurs anywhere in the LNM.

The etalon is a flat parallel plate. It does not focus, expand, or collimate. The beam exits the same diameter it entered. The slight tilt from the PZT is so small (microradians) that any lateral displacement is negligible.

Since the etalon is a filter, and filtering costs power - two losses occur:

Loss mechanism	Cause	Typical magnitude
Reflective loss	The partial-R coatings reflect some of λ_0 back rather than transmitting it	10 – 40% per pass depending on coating R
Bandwidth rejection	All power outside the sub-pm passband is reflected back and eventually lost from the cavity	The raw bandwidth is narrowed ~1000×, so the vast majority of spontaneous emission energy never builds up

The net result is that the output beam carries significantly less power than the gain medium theoretically could produce. This is accepted as the necessary cost of achieving the spectral purity that the projection lens requires. In practice DUV excimer lasers compensate by running at high pulse repetition rates (1–4 kHz) and high pulse energies rather than by making the LNM more transmissive.



Working Principle

The transmission condition is:

```
2 × n × d × cos(θ) = m × λ
```

where

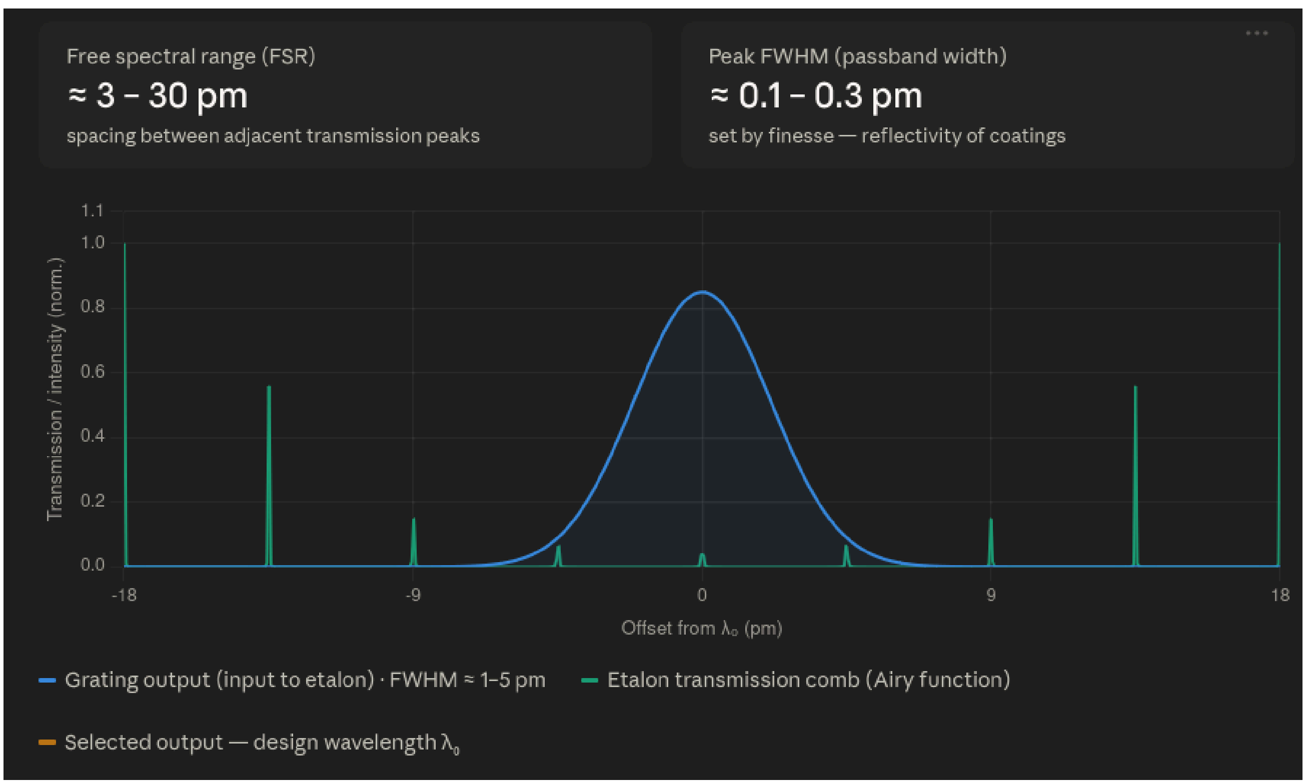
- n is the refractive index of the plate,
- d is the plate thickness,
- θ is the angle of incidence,
- m is an integer (the interference order).

Only wavelengths that make the round-trip path length an exact integer multiple of themselves add up constructively and pass through. Everything else cancels out by destructive interference.

Parameter Changed: Spectral Bandwidth (Ultra-Fine Narrowing)

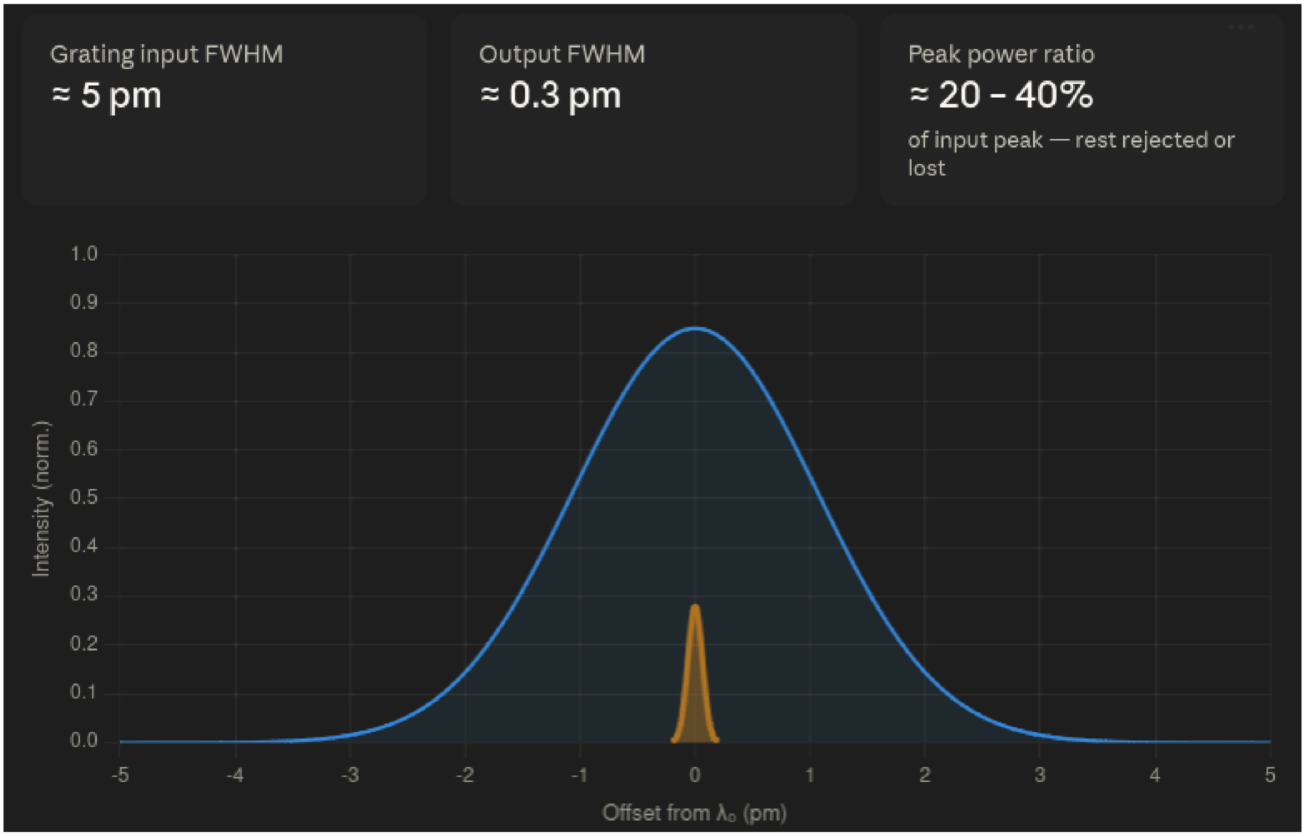
- Before (Grating Output):** The coarsely narrowed spectrum (tens of picometers wide), which still contains slight variations in color that would cause micro-blurring in a projection lens.

- After (Etalon Output):** The final, ideal lithography spectrum. The output is compressed into a razor-thin, ultra-pure spectral line (e.g., <0.3 pm FWHM), stripping away all remaining spontaneous emission and leaving only a single, perfectly pure wavelength.



Tilting the etalon plate slightly changes the optical path length 2·n·d·cos θ.

This slides the entire green comb left or right along the wavelength axis by a few pm — moving the central peak precisely onto λ_0. This is the fine-tuning mechanism. The wavemeter downstream monitors where the output peak sits and feeds back to the PZT to keep it locked.



Blue curve — the grating output (what feeds into the etalon) This is the spectrum arriving at the etalon after the echelle grating has already done its coarse narrowing. It is a smooth broad hump roughly 1–5 pm wide, centred at λ_0 (zero on the x-axis). This is still far too wide for the projection lens — every wavelength within that hump would cause chromatic aberration.

Green comb — the etalon's transmission function (the Airy function) This is the key. The etalon does not have a single passband — it has a periodic series of razor-thin transmission peaks, equally spaced by the Free Spectral Range (FSR). In the chart the FSR is ~9 pm, so peaks appear at −18, −9, 0, +9, +18 pm and so on. Each peak is only ~0.1–0.3 pm wide. Between the peaks, transmission drops to nearly zero — those wavelengths are reflected back. The comb shape is described mathematically by the Airy function, which comes directly from the multi-beam interference physics inside the etalon plate. Higher finesse (higher coating reflectivity) makes the peaks sharper and the valleys deeper.

Amber filled region — the actual laser output This is what you get when you multiply the two curves together — the blue input spectrum multiplied by the green transmission comb. Only where a green peak coincides with the blue hump does any light get through. The grating ensures only one green peak sits under the blue hump — the one at λ_0 (zero). The result is the amber spike: a single sub-pm passband, the final narrowed output of the LNM.

Rejected wavelengths

The rejected wavelengths do not exit — they are reflected back from inside the etalon and re-enter the main laser cavity. The sequence is:

Light enters the etalon from the gain medium side (right face). At the left face, the partial-R coating reflects most of it back. The reflected beam bounces between the two etalon surfaces multiple times. Because the round-trip path length is not an integer multiple of λ, all the internal reflections arrive out of phase and cancel each other destructively. The net result is that the etalon appears as a near-perfect mirror for those wavelengths — almost all energy is sent back into the main cavity toward the gain medium and grating.

So rejected wavelengths traverse the cavity path again, re-enter the gain medium, get re-amplified, and re-compete — but they keep losing at the etalon on every round trip and never build up.

Key Spectral Parameters

The following are the spectral parameters tuned in the Elaton.

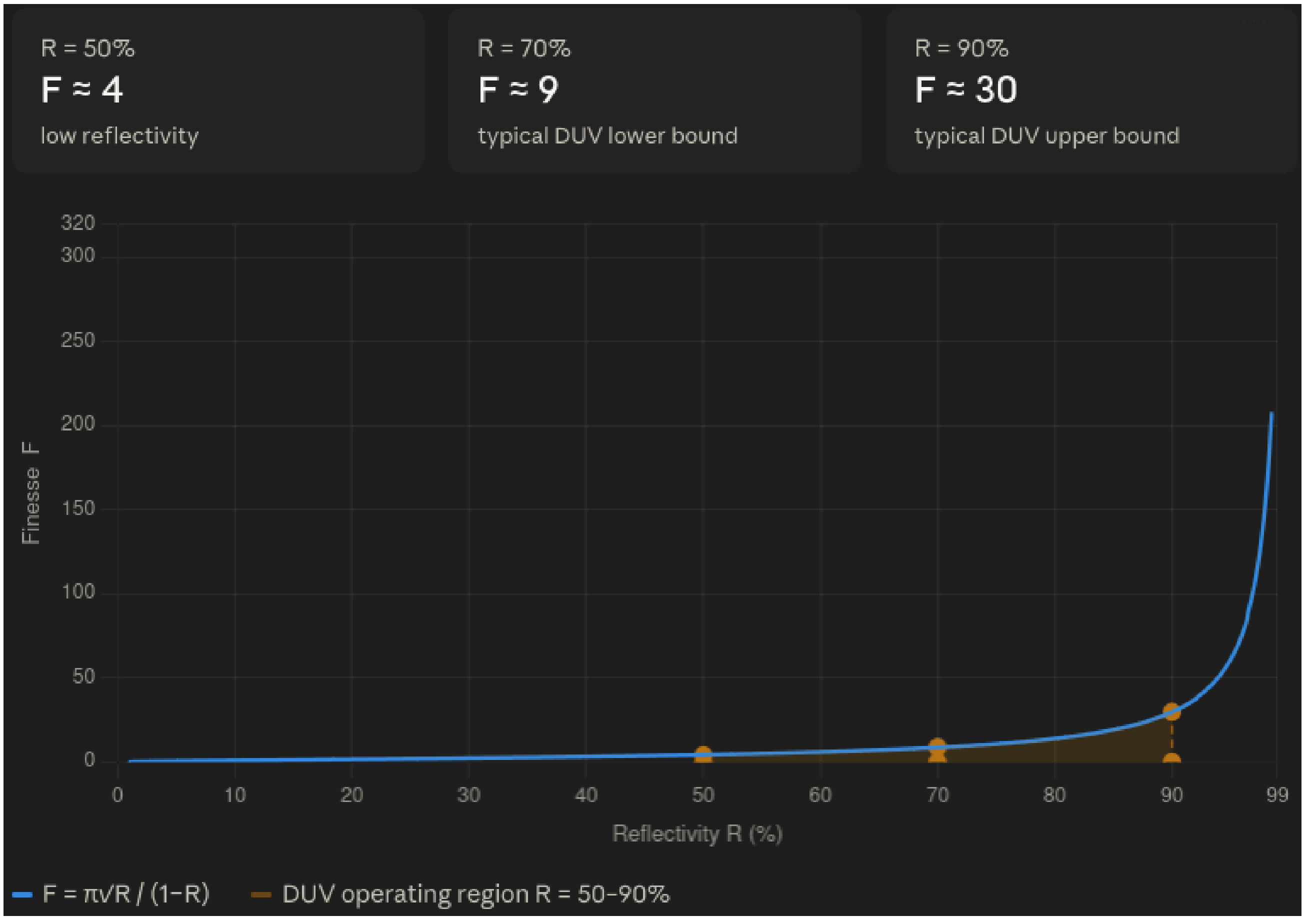
Parameter	Definition	Typical DUV Value
Plate thickness `d`	Physical thickness of the etalon	1 – 10 mm
Coating reflectivity `R`	Reflectance per surface	50 – 90%
Finesse `F`	FSR / FWHM per peak ≈ $\pi\sqrt{R} / (1 - R)$	30 – 100
Free spectral range (FSR)	Spacing between adjacent transmission peaks	3 – 30 pm
Passband FWHM	Width of each transmission peak	0.1 – 0.3 pm
Tuning mechanism	PZT actuator on plate tilt or gap	± few pm

Finesse

Finesse is the single most important figure of merit. It is set entirely by the coating reflectivity:

```
F = π × √R / (1 − R)
```

Higher R → higher finesse → narrower, sharper transmission peaks.



Finesse vs Power Trade-off in # Fabry-Pérot Etalon

Power and finesse pull in opposite directions — they are a direct trade-off set by the same coating reflectivity R.

The peak transmission of the etalon at λ_0 is:

$$T_{peak} = (1 - R)^2 \,/\, (1 - R)^2 = 1 \qquad \text{(ideal, lossless coatings)}$$

But real coatings absorb and scatter a small fraction A of light on each reflection. The actual peak transmission becomes:

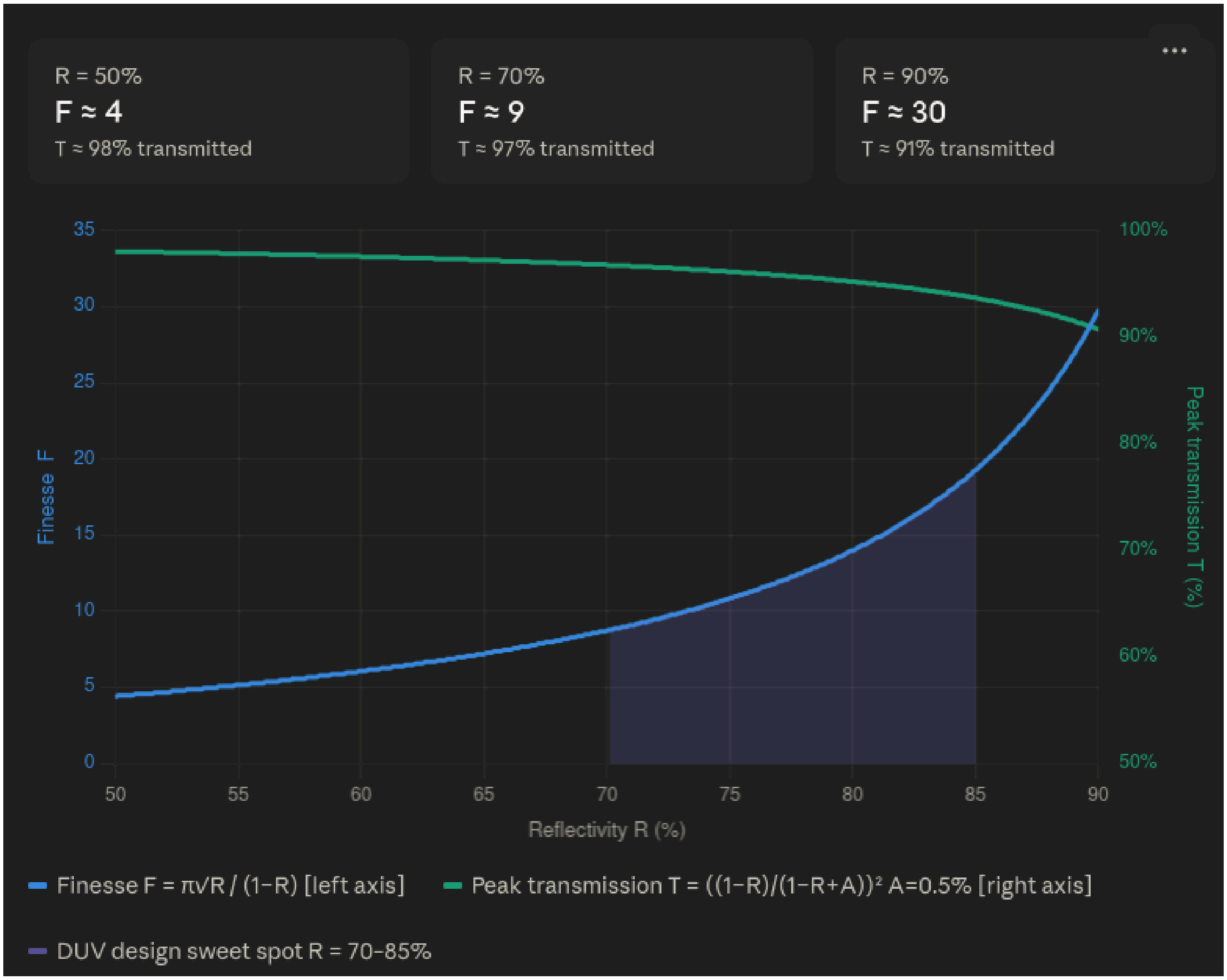
$$T_{peak} = \left((1 - R) \,/\, (1 - R + A) \right)^2$$

Effect of Increasing R

R increases	Effect on finesse	Effect on peak transmission
R → higher	F = π√R/(1−R) → rises steeply	T_peak → falls — less power exits
R → lower	F → falls — broader peaks	T_peak → rises — more power exits

Key Values in the DUV Operating Region (R = 50–90%)

R	Finesse F	Peak transmission T	Consequence
50%	≈ 9	≈ 98%	Nearly all power exits but peaks are broad — poor selectivity
70%	≈ 18	≈ 88%	Moderate finesse, good transmission — practical balance
90%	≈ 30	≈ 55%	Sharp peaks, but nearly half the power is lost at the etalon



Finesse vs Power Design Sweet Spot

DUV laser manufacturers choose R in the **70–85% range** as the engineering sweet spot:

- High enough finesse to hit the sub-pm bandwidth specification
- Low enough reflection loss that the wafer dose budget can still be met

The coating absorption A ≈ 0.5% means transmission never quite reaches 100% even at very low R — every reflection loses a small fraction to heat regardless of wavelength.

Formulas

```
Finesse:              F = π × √R / (1 − R)

Peak transmission:    T = ( (1 − R) / (1 − R + A) )²

where:
  R = coating reflectivity (per surface)
  A = coating absorption loss per reflection (typically 0.3–0.8% for
DUV coatings)
```

The single parameter R controls both quantities simultaneously and in opposite directions. There is no configuration that gives both high finesse and high transmission — every DUV LNM design is a deliberate compromise between spectral purity and output power.

Ref -

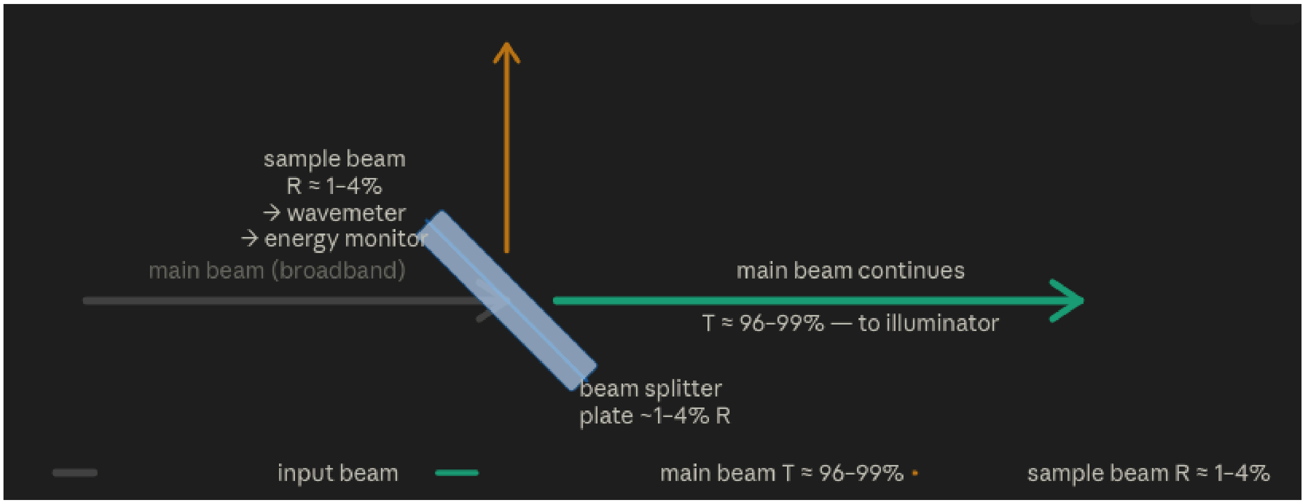
https://optics.ansys.com/hc/en-us/articles/360042323754-Fabry-Perot-cavity-INTERCONNECT

3.4 Bandwidth & Wavelength Metrology (In-Line Monitor)

This stage does not process the beam — it watches it. A small fraction is continuously sampled and measured in real time. The three instruments run simultaneously on every pulse, typically at 1–4 kHz, and their outputs close the feedback loop back to the LNM.

Component	Function
Beam Splitter Pick-off	Samples a small fraction (~1–4%) of the laser output without perturbing the main beam.
Wavemeter / Spectrum Monitor	A compact grating-based or etalon-based spectrometer that measures **center wavelength (λ_c), bandwidth (E95 or FWHM), and pulse energy** in real time. Feeds back to the LNM tuning actuators.
Energy Monitor (pyroelectric detector)	Measures pulse-to-pulse dose for feedback to the attenuator.

Step 1 - Sample Output - Beam Splitter Pick-off

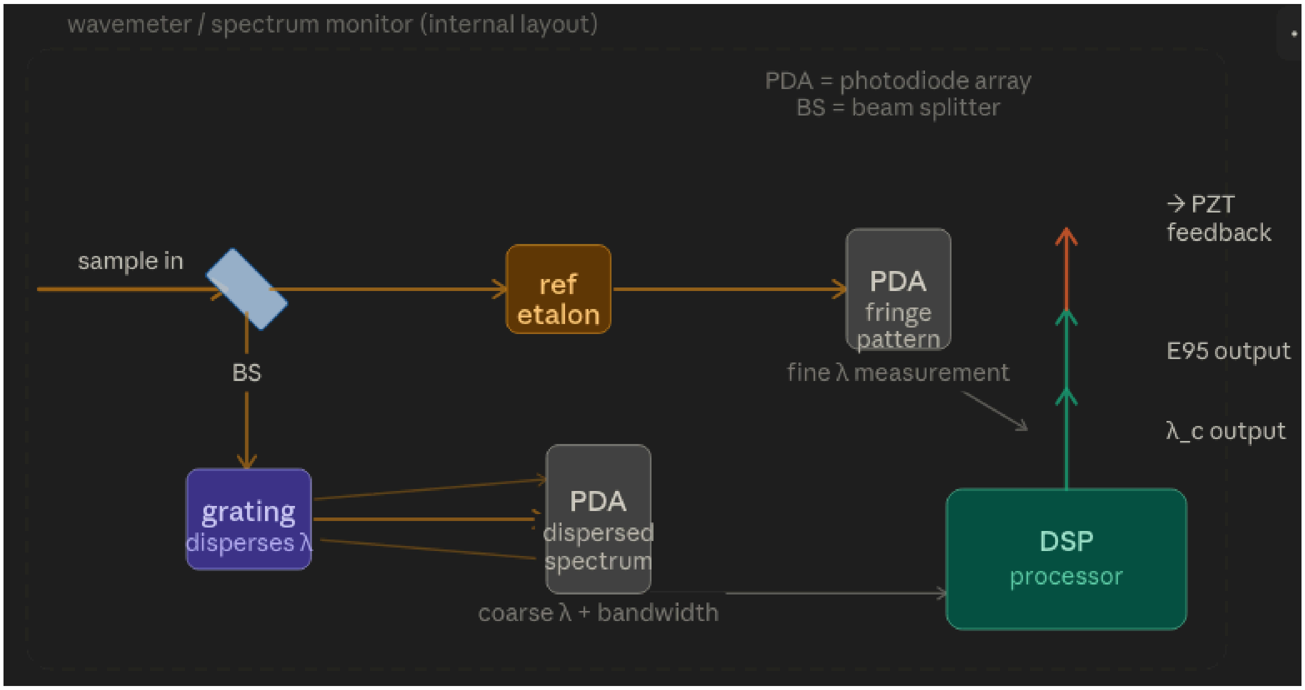


The pick-off is a fused silica or CaF_2 plate tilted at a small angle to the beam (typically 1–5°, not the 45° shown — the diagram exaggerates for clarity). The coating is designed to reflect just 1–4% of the incident beam upward into the metrology branch, while transmitting 96–99% straight through to the illuminator. The plate must be:

- DUV-grade fused silica or CaF_2 — any other glass absorbs at 193/248 nm
- Wedged slightly to prevent the back-surface reflection from creating a second ghost beam that would confuse the wavemeter
- AR-coated on the back surface to suppress that ghost entirely
- Temperature-stabilised — thermal expansion changes the angle and shifts the sampled fraction

The pick-off does not perturb the main beam in wavelength or spatial coherence. The only effect is a ~1–4% power reduction, which is budgeted into the dose calculation.

Step 2a - Digitize Physical Parameter - Wavemeter / Spectrum Monitor



The wavemeter uses two parallel measurement paths on the same sample beam:

Path 1 — Reference etalon + photodiode array (PDA): The sample beam passes through a temperature-stabilised reference etalon. The transmitted fringe pattern falls on a linear PDA. Because the etalon's FSR is precisely known and thermally locked, the fringe spacing gives the centre wavelength λ_c to ±0.01 pm resolution. This is the fine measurement.

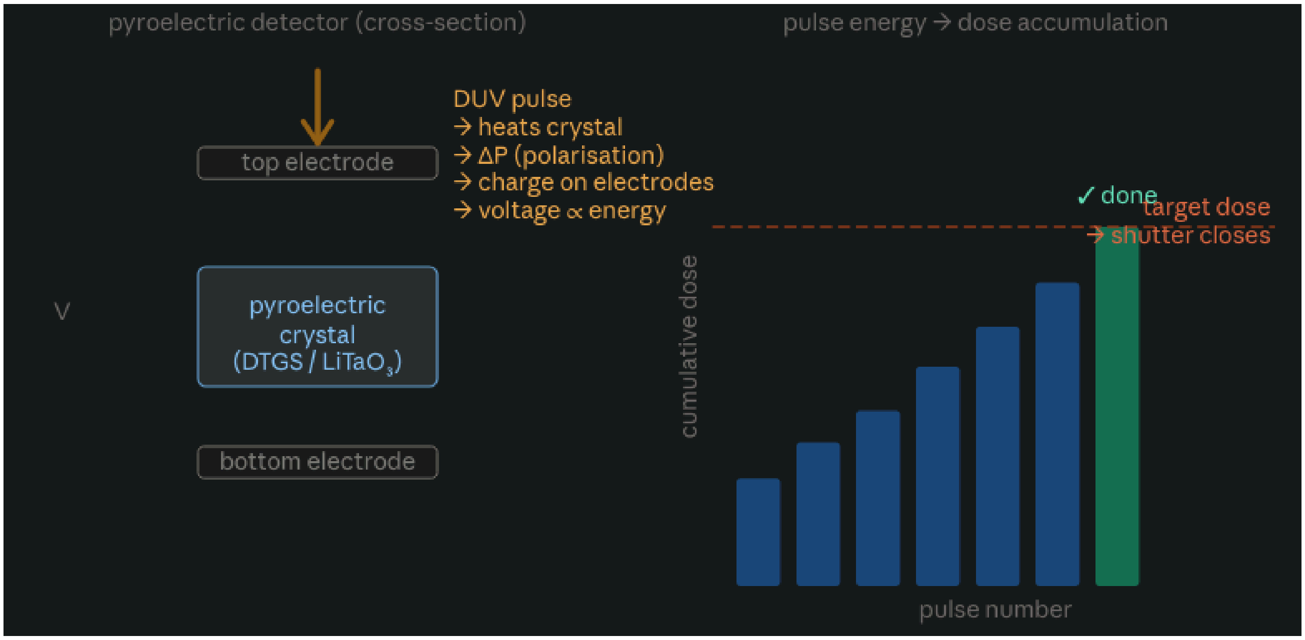
Path 2 — Diffraction grating + photodiode array: A small grating disperses the beam angularly across a second PDA. The spatial spread of the detected spectrum directly gives the bandwidth (E95 — the spectral window containing 95% of pulse energy). This is the coarse measurement but captures the full spectral shape.

Both arrays are read out on every laser pulse by the DSP processor, which:

- Computes λ_c and E95 within microseconds
- Compares λ_c to the target wavelength set point
- Sends a correction signal to the LNM PZT actuator to pull λ_c back on target
- Logs every pulse to the scanner controller for dose accounting

The reference etalon inside the wavemeter is temperature-controlled to ±0.001°C because its FSR is the absolute wavelength reference — any thermal drift would shift every subsequent measurement.

Step 2b - Digitize Physical Parameter - Energy Monitor (Pyroelectric Detector)



The pyroelectric detector measures energy per pulse, not wavelength. The operating principle:

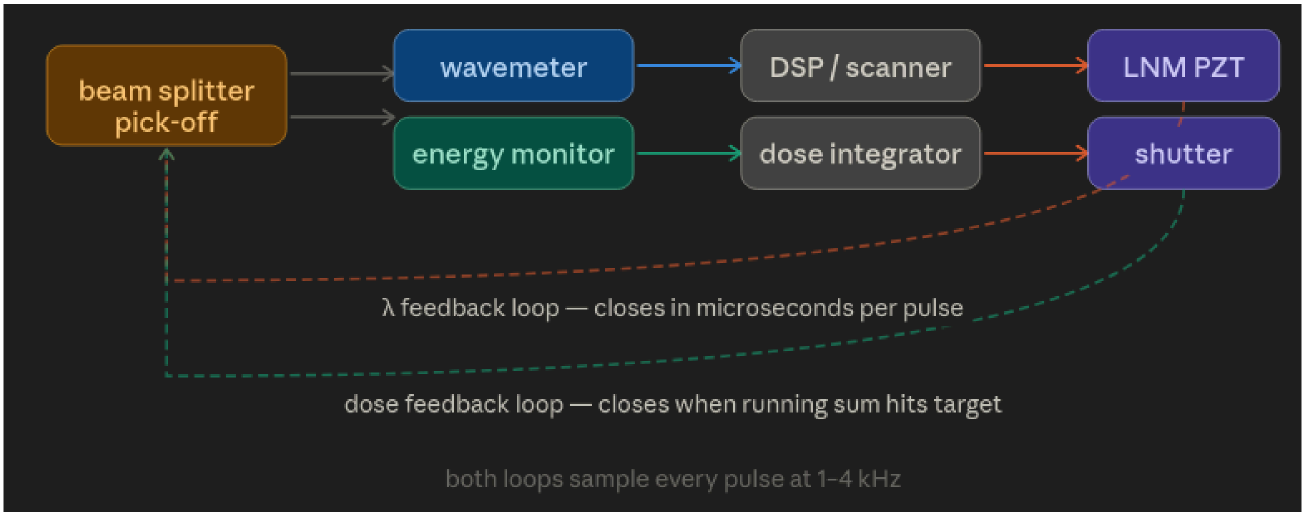
1. A DUV-absorbing crystal (typically DTGS — deuterated triglycine sulphate — or $LiTaO_3$) absorbs the incoming pulse and heats slightly — a temperature rise of microkelvin per pulse.
2. The heating changes the crystal's internal electric polarisation (the pyroelectric effect).
3. This polarisation change drives a charge pulse onto the two electrodes sandwiching the crystal.
4. The charge pulse is integrated by a transimpedance amplifier into a voltage proportional to the pulse energy.
5. The voltage is digitised and logged by the scanner controller.

How it controls dose: The scanner controller maintains a running sum of pulse energies delivered to the current exposure field. When that sum reaches the target dose (mJ/cm^2), the shutter closes — stopping further pulses from reaching the wafer. Because each pulse is measured individually, the system compensates for pulse-to-pulse energy variation (which can be ±5–10% in a raw excimer) by adjusting how many pulses are fired per field.

Why pyroelectric and not a photodiode: At 193 nm, conventional silicon photodiodes degrade rapidly under DUV irradiation. Pyroelectric detectors are thermal devices — they respond to heat, not photons directly — making them insensitive to DUV-induced damage. They also have a very flat spectral response, so the reading does not depend on wavelength.

Step 3 - Feedback Control Loop

The three components above form two closed feedback loops:



Loop 1 — Wavelength control (red): Wavemeter → DSP → LNM PZT. Closes in microseconds. Keeps λ_c locked to target within ±0.01 pm on every pulse.

Loop 2 — Dose control (green): Energy monitor → dose integrator → shutter. Closes when the cumulative dose sum reaches the exposure target. Compensates for pulse-to-pulse energy variation so the wafer always receives the correct total dose regardless of individual pulse fluctuations.

The beam splitter pick-off is the common entry point that feeds both loops simultaneously from a single unperturbed sample of the main beam. Shall I export this as a markdown file?

3.5 — Beam Conditioning & Delivery Unit (BDU)

The BDU receives the spectrally pure, metered beam on a a narrow, near-Gaussian spatial profile from the metrology stage and prepares it physically for the illuminator.

Three problems must be solved - the beam is:

- too narrow
- too coherent
- too energetic

Each component fixes one of those.

Component	Function
Variable Attenuator	A pair of partially reflective plates (or a rotating waveplate + polarizer) that continuously varies transmission to control **dose on wafer** without changing laser operating point.

Component	Function
Beam Shaping / Expanding Telescope	Refractive or reflective telescope that expands the ~1×3 cm excimer beam to fill the illuminator entrance pupil.
Coherence Scrambler / Light Pipe	A quartz or CaF_2 **kaleidoscope rod** or **fiber bundle** that scrambles spatial coherence. This reduces laser speckle and prevents interference artifacts in the aerial image.

Each component fixes exactly one property of the beam — power, size, coherence, and spectral purity — without affecting the others. By the time the beam reaches the illuminator entrance, it is the correct energy, the correct diameter, spatially uniform, low coherence, and spectrally pure.

3.5.1 Variable Attenuator

The attenuator controls how much energy reaches the wafer without touching the laser operating point. If you reduced dose by running the laser at lower power, the discharge plasma chemistry would change — affecting wavelength stability and bandwidth. Instead the laser runs at its optimal operating point always, and the attenuator dials the beam down to whatever the exposure recipe requires.

How it works: A rotating half-wave plate (HWP) followed by a fixed polariser. The HWP rotates the beam's polarisation angle by 2θ. The polariser transmits only the component aligned with its axis — blocking the rest. Transmission follows Malus's law:

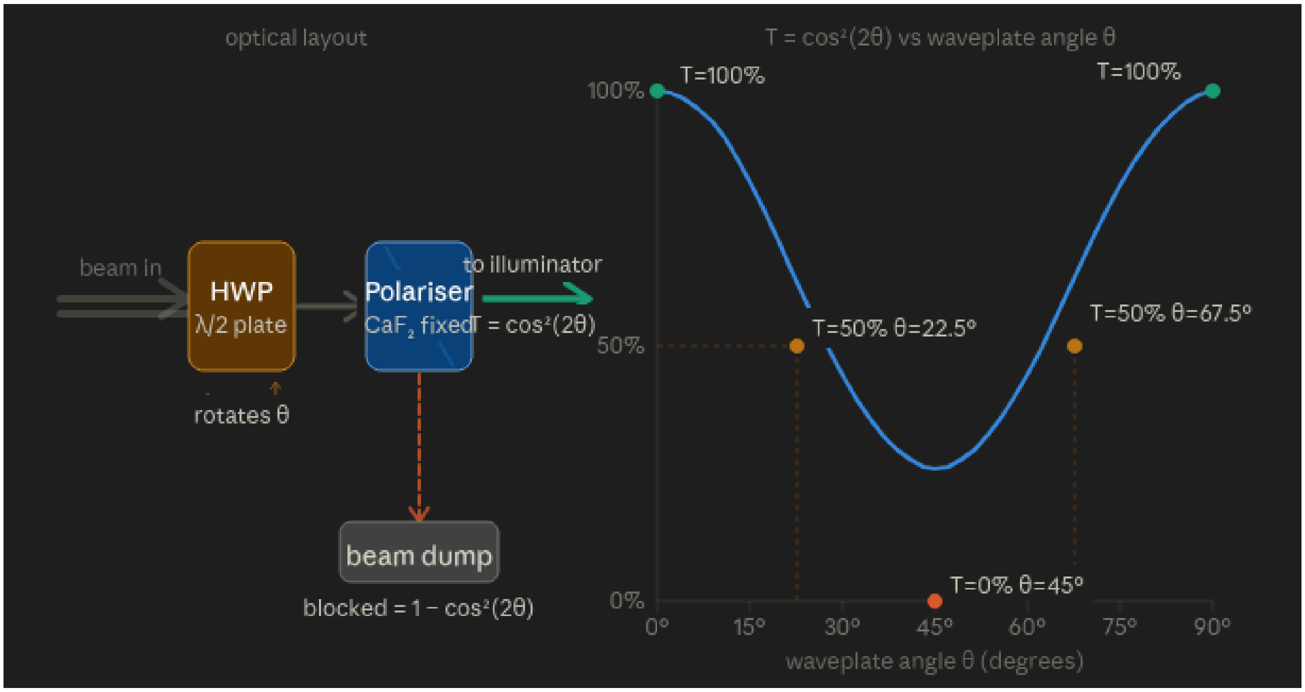
```
T = cos²(2θ).
```

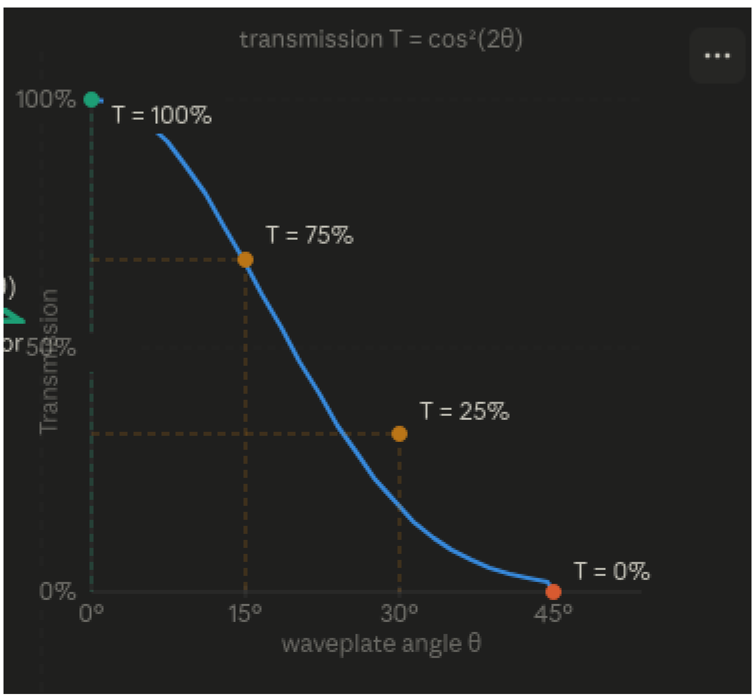
Where

- T - Light intensity
- If you rotate the HWP by $\theta=15\circ$, the exit light's polarization rotates by $30\circ$.
- If you rotate the HWP by $\theta=45\circ$, the exit light's polarization rotates by $90\circ$

Rotating the HWP by 45° takes transmission from 0% to 100% continuously and smoothly.

The blocked component is not absorbed by the polariser — it is reflected or deflected away as a beam dump. The CaF_2 polariser must survive the full DUV irradiance, and the beam dump must safely absorb it without generating back-reflections.





Fine control — the attenuator The attenuator provides fine, continuous, real-time trimming of the pulse energy to hit the exact target dose. Specifically it solves three problems coarse control cannot:

1. Partial pulses at field edges As the reticle scan starts and ends, the illuminated field ramps in and out. The last pulse before the shutter closes may only partially expose the field. The attenuator can reduce that final pulse's energy precisely so the cumulative dose lands exactly on target rather than overshooting.

2. Pulse-to-pulse energy variation The excimer laser has inherent shot-to-shot energy variation of ±5–10%. The energy monitor (Stage 2) measures each pulse, and the attenuator adjusts on the very next pulse to compensate. This closed-loop correction keeps dose uniformity within ±0.5% across the field.

3. Dose recipe changes between layers Different process layers require different doses (mJ/cm²). Rather than retuning the laser for each layer, the attenuator is simply rotated to a new angle. The laser stays at its operating point throughout.

It is not a safety shutter — a separate fast mechanical shutter handles that. The attenuator is a precision dose control tool. The analogy is a dimmer switch on a light — the power station (laser) keeps running at full capacity; the dimmer (attenuator) controls what reaches the room (wafer).

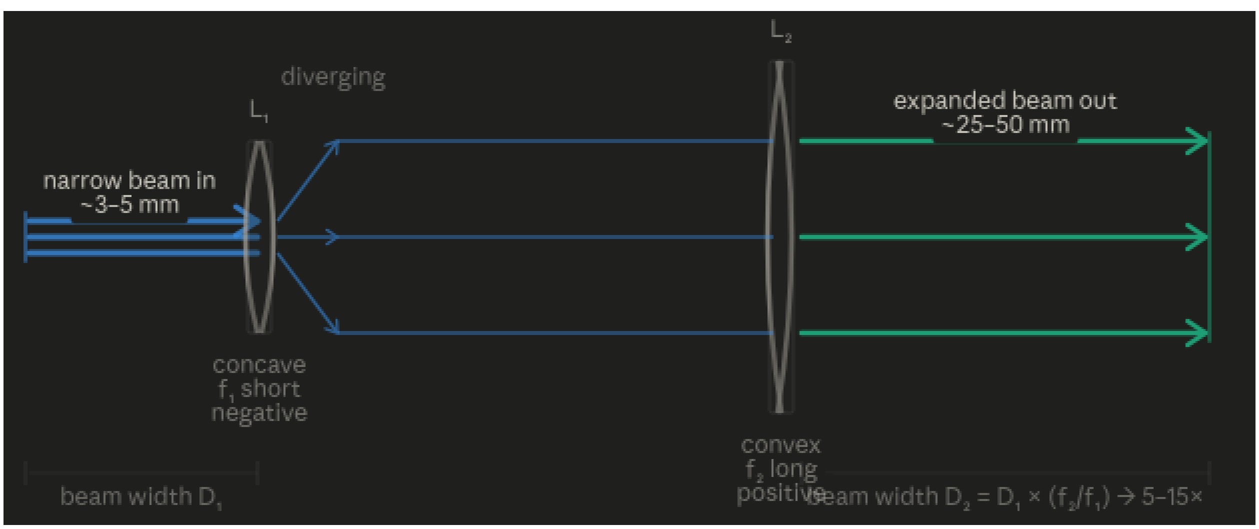
3.5.2 — Beam Expanding Telescope

The excimer beam exits the LNM as a narrow ~3–5 mm ribbon. The illuminator entrance pupil needs a beam ~25–50 mm wide to fill the fly's eye lens arrays. The telescope solves this with two lenses:

L_1 — a short focal length concave (negative) lens that diverges the beam.

L_2 — a long focal length convex (positive) lens placed one focal length downstream that recollimates the diverging beam into a wider parallel beam.

The expansion ratio is simply f_2/f_1.



Both lenses must be fused silica or CaF_2 — no other optical glass transmits at 193 nm. The lenses are also AR-coated for DUV to minimise reflection losses at each surface. Because the beam is now expanded, the irradiance (W/cm²) at the lens surfaces is lower — this is intentional, as high irradiance would cause photochemical compaction damage in the glass over time.

3.5.3 — Coherence Scrambler / Light Pipe

The beam expander magnifies whatever spatial structure exists in the laser output. The excimer discharge is not perfectly homogeneous; it fires across a rectangular electrode gap and the plasma density varies slightly across the aperture. The prism beam expander stretches this non-uniformity further.

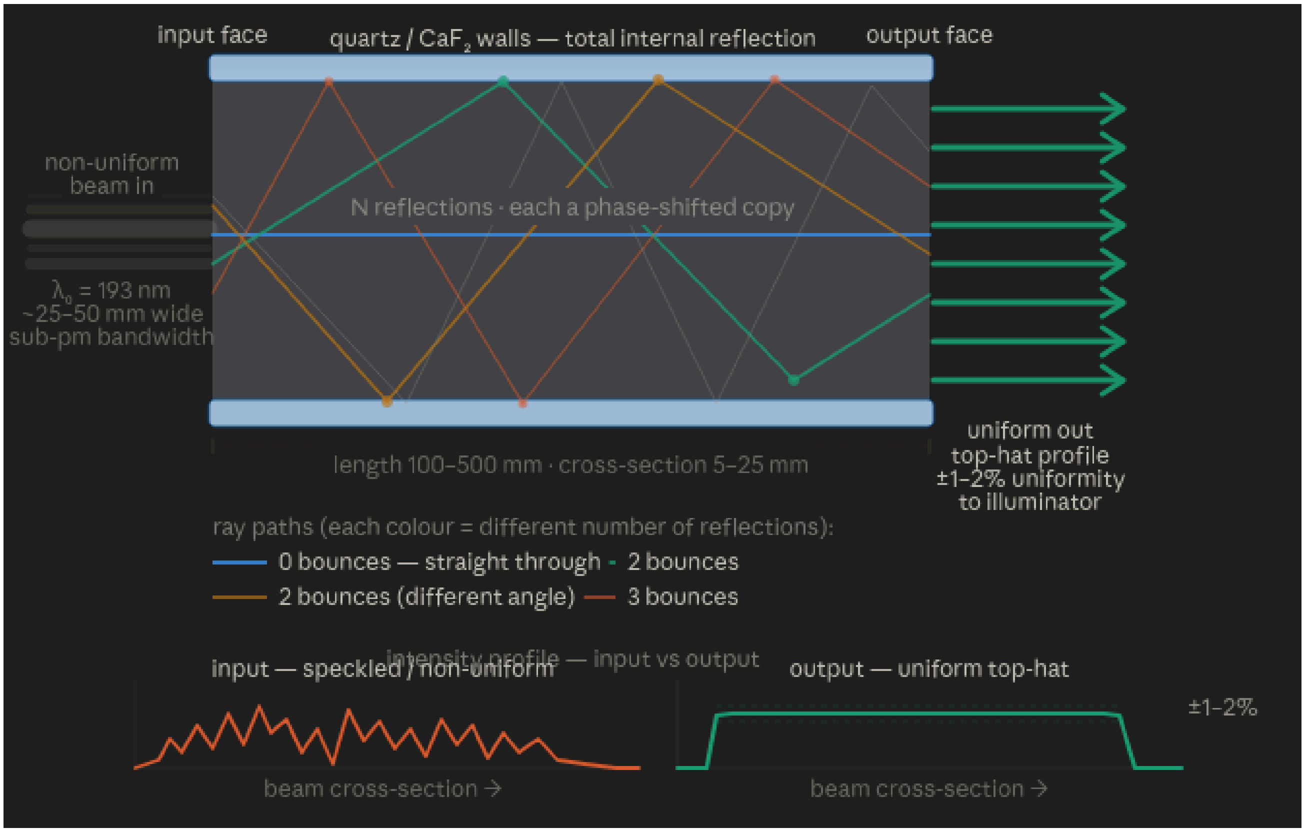
The result is a 25–50 mm wide beam that may have:

- Bright bands aligned with the electrode geometry
- Gaussian-like roll-off toward the beam edges
- Interference fringes from residual spatial coherenc

The light pipe is a solid rectangular rod of polished fused silica or CaF_2, typically 100–500 mm long with a 5–25 mm square cross-section. The beam enters the input face and undergoes hundreds of total internal reflections off the four polished walls as it propagates through. Each reflection creates a laterally displaced, phase-shifted copy of the input beam. All these copies superimpose at the output face, averaging out:

- Each reflection off a wall creates a laterally displaced copy of the input beam.
- After N reflections: N copies, each shifted by a different amoun and all superimposed at the output face.
- Any bright spot in the input appears N times across the full aperture → averages out.
- Any dark spot is filled in by bright copies from adjacent regions → averages out.

The light pipe is entirely passive. There are no sensors, no actuators, no control loop. The light pipe cannot be "too slow" or "lose lock" because it requires no lock. It simply works on every pulse by the same physical law — total internal reflection — that has no time constant



All these copies superimpose at the output face, averaging out:

- Hot spots — bright regions of the input are spread across the full aperture
- Speckle — interference fringes are washed out because each copy has a different phase delay
- Spatial coherence — neighbouring points on the output face come from entirely different reflection paths, so they carry no fixed phase relationship

The output is a spatial top-hat — uniform intensity across the full aperture — which is exactly what the illuminator fly's eye arrays need as their input. The light pipe does not change wavelength, polarisation state (on average), or total power — only the spatial distribution and coherence.

Ref -

https://www.researchgate.net/publication/269322335_The_influence_of_a_light_pipe_on_the_coherenc

https://kuleuven.limo.libis.be/discovery/fulldisplay?docid=lirias867703&context=SearchWebhook&vid=32KUL_KUL:Lirias&lang=en&search_scope=lirias_pr

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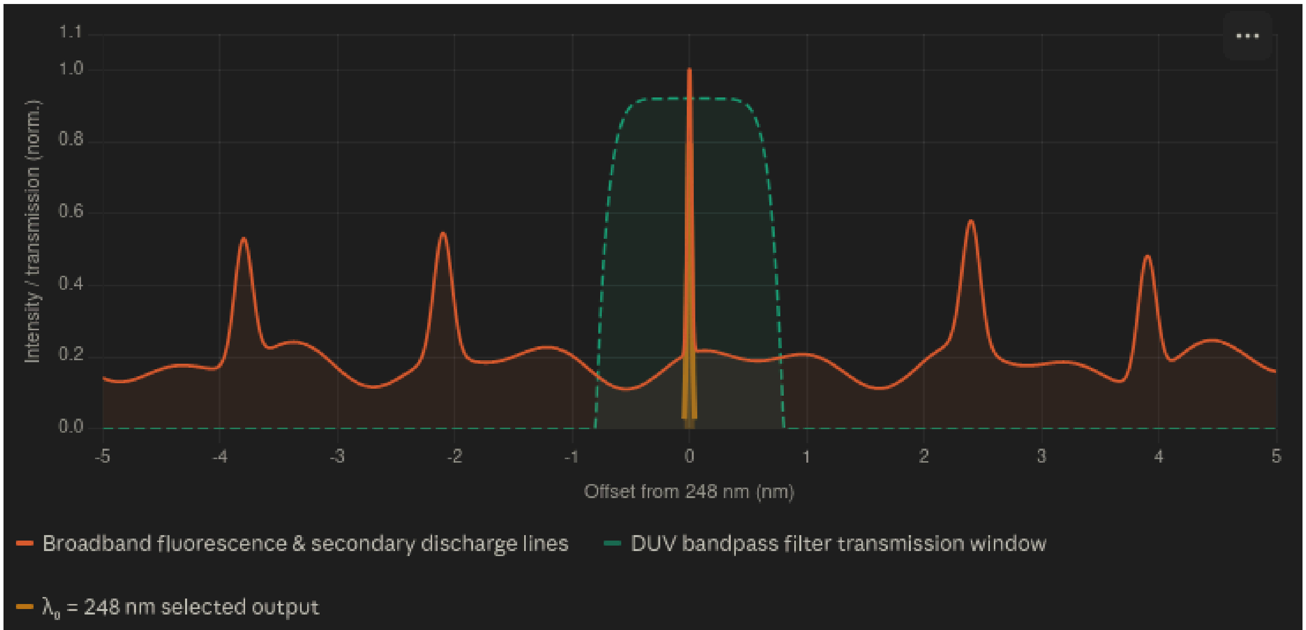
3.5.4 — DUV Bandpass Filter (KrF systems)



Note: x-axis spans ±5 nm around 248 nm. Broadband fluorescence and secondary KrF plasma lines outside the filter window are rejected before the illuminator.

The DUV bandpass filter is present on KrF (248 nm) systems only — ArF at 193 nm does not require it because the ArF plasma produces a cleaner spectrum. KrF discharges generate:

- Broadband fluorescence — a continuous low-level background across a wide wavelength range from excited Kr and F_2 species
- Secondary discharge lines — additional atomic emission lines from Kr and F at wavelengths away from 248 nm

These would pass through the LNM, enter the illuminator, and reach the wafer where they would expose the resist at the wrong wavelength — causing dose errors and CD variation. The bandpass filter is a multi-layer dielectric interference coating on a CaF_2 substrate, centred at 248 nm with a passband of ~0.5–1 nm. Everything outside is reflected back and lost. The filter has no moving parts and requires no feedback — it is passive and fixed.

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